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"Factors affecting the intention to use artificial intelligence in mobile banking"

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Graduation project 2 Report

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In the Name of Allah, the Most Merciful, the Most Compassionate, all praises belongs to Almighty Allah, the Lord of the worlds and prayers and peace be upon Muhammad His servant and messenger.

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Abstract

This study examines the adoption of artificial intelligence (AI) in mobile banking, focusing on four main dimensions which is Trust, User Satisfaction, perceived ease of use and Use Behavior.

The goal of this project is to improve the experience of artificial intelligence for customers and to demonstrate the capabilities of artificial intelligence in the decision-making process and financial services, as many customers have concerns about sharing their information with artificial intelligence and how to deal with it, so this will highlight the possibilities and ease of artificial intelligence. The results provide valuable information for financial institutions that want to improve AI implementations and build user confidence and satisfaction in mobile banking

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CHAPTER ONE:

INTRODUCTION

CHAPTER ONE

INTRODUCTION

1.1 Background

Technology has a very big impact on our daily lives and it is impossible to visualize a life devoid of the technology that makes our life easier and six billion and it is expected to grow further by several hundred million

in the next few years. Examples of technology include mobile devices and artificial intelligence. Therefore, we will see the extent of the impact of artificial intelligence on m-banking. This is what we will talk about today: “The intention to use artificial intelligence in mobile banking.” (3)

M-banking refers to mobile banking, also known as m-banking, which enables customers to communicate directly with their bank and conduct financial transactions through mobile devices such as smartphones and tablets. (2)

With the emergence of these programs, it has become easier to use banking services.

Some of these services include viewing your balance, paying bills, transferring money, and purchasing from websites or stores, whether products or services, at any time and in any place via electronic devices. (1)

Therefore, there are many factors that negatively affect the adoption and success of mobile banking services. Fear of electronic transactions, fraud, lack of internet and other technological means are important reasons for not adopting artificial intelligence. (3)

and Ease of use, performance, customer trust, satisfaction and user behaviour are key factors that positively impact mobile banking if used correctly.

As banking institutions integrate advanced technologies such as artificial intelligence, understanding these factors becomes more important in driving mobile banking adoption and improving customer experiences.

Understanding the factors that influence customer satisfaction has become very important for financial institutions seeking to retain and attract users because customer satisfaction is one of the key factors determining success in the banking industry as it directly impacts customer loyalty, retention, and overall profitability. (3)

This study aims to understand customer behavior and facilitate the use of the program in order to enhance customer trust and satisfaction and reduce their concerns about artificial intelligence.

1.2 Problem statement

The adoption of artificial intelligence in mobile banking faces Important challenges, despite its potential to improve customer experience, assist security and streamline banking operations. Many banking institutions are make a Serious mistake to integrate AI technologies due to data privacy concerns, lack of user trust, and the complexity of AI systems. In addition, customers may be reluctant to adopt AI-based features such as chatbots, automated financial advice or biometric evidence

because of lack of awareness, risk perception or fear of security breaches. This study aims to explore the factors hold up the widespread adoption of AI in mobile banking and identify strategies to overcome these barriers to ensure smooth and secure user experiences.

1.3 Study Importance

The importance of Intention to Use AI in Mobile Banking is in some concepts, Including User adoption which means Understanding customer intent to use AI in mobile banking helps banks tailor services to user needs, thus driving faster adoption.

Also Improved customer experience, When customers aim to use AI, banks can focus on features such as support and personalized recommendations, improving satisfaction and loyalty.

Moving on to Effective service delivery, A strong commitment to using AI allows banks to confidently invest in automation, improving services such as customer support, fraud detection and transaction processing.

also Increased security Customers who intend to use AI-based security features (e.g. fraud detection, biometric identification) contribute to a safer banking environment.

Down to Competitive advantage, The positive intention to use AI drives innovation, placing banks ahead in a technology-driven market and attracting technological customers.

1.4 Research question

Based on the above discussion, this study sought to answer the following questions:

- What are the factors that affect Intention to use artificial intelligence in mobile banking?
- To what extent does each factor affect the adoption of artificial intelligence in banking applications from the users' perspective?

1.5 Research objectives

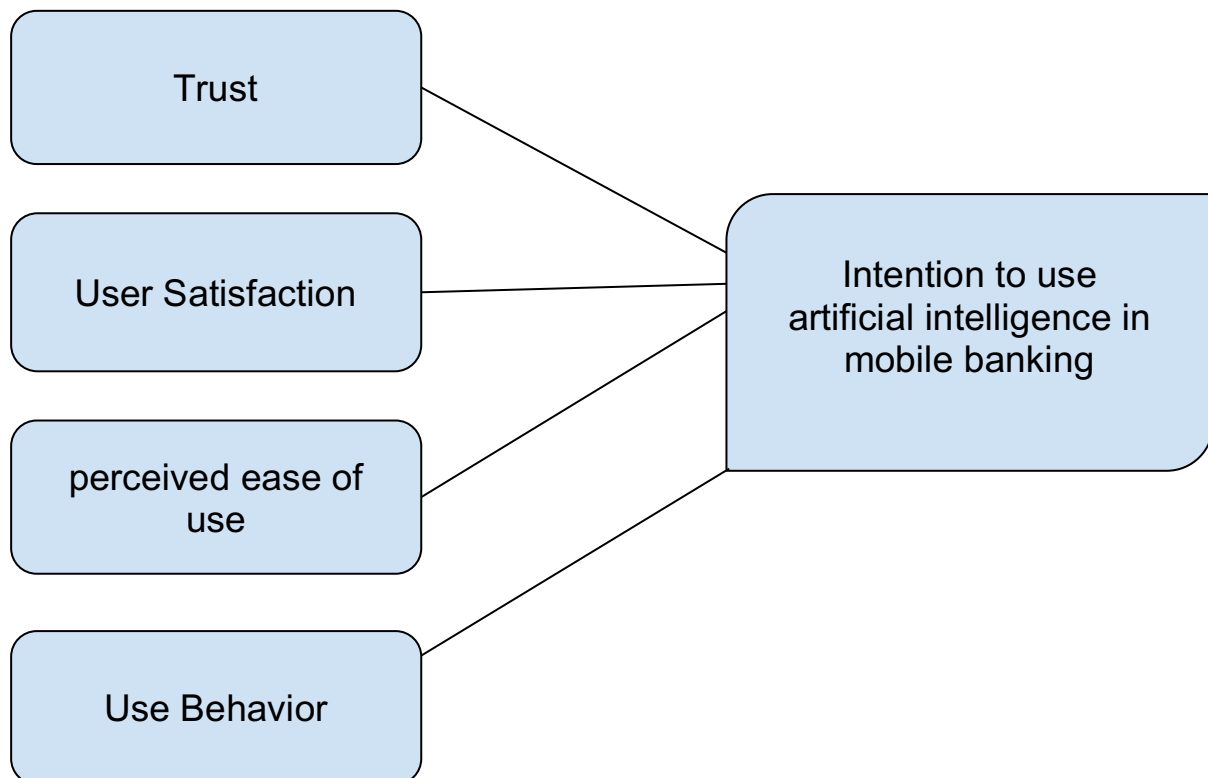
The main objective of the current study is to explore the factors that influence the intention to adopt the use of artificial intelligence in mobile banking from the users point of view,

and this will be achieved through the following objectives:

- Literature review to provide a strong theoretical background regarding
An intention to adopt the use of artificial intelligence in mobile banking
- Develop and examine a research model to study the factors affecting it.

1.6 Research Model

This Figure 1.1 well represent research model (1),(2),(3)



CHAPTER TWO:

Theoretical Framework

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Theoretical Framework

This chapter provides the theoretical foundations on which our research is based. Including the following question : What are the factors affecting Intention to use artificial intelligence in mobile banking? How the factors can affect? We also provided an explanation of the concepts of the basic influential elements, alongside the adoption of artificial intelligence and its adoption in mobile banking.

2.1 Definitions:

ARTIFICIAL INTELLIGENCE

(AI) Seeks to make computers do the sorts of things that minds can do . (7)

Mobile banking

also known as m-banking, refers to performing financial transactions through mobile devices such as smartphones and tablets. (8)

Trust

Trust is the belief or confidence that someone or something is reliable, honest, and will act in your best interest. (3)

User satisfaction

is the feeling of happiness or fulfillment a user has when a product, service, or experience meets their needs and expectations. It reflects how well the product works for them and how enjoyable or effective they find it. (1)

Perceived ease of use

is how simple and effortless a person believes it will be to use a product or system. If something is easy to understand and operate without much effort, it has a high perceived ease of use.(1)

Use Behavior

is the way people actually interact with and use a product or system in real life. It shows how often, how long, and in what ways the product is used (2)

2.2 Intention to use artificial intelligence in mobile banking

First, let us know what artificial intelligence is refers to the simulation of human intelligence in machines programmed to perform tasks that typically require human cognitive abilities, such as learning, problem solving, reasoning, understanding the language and perception. AI systems can adapt to new inputs and improve their performance over time using machine learning and data-driven models.

AI was officially recognized as a field of study in 1956 at the Dartmouth Conference, organized by computer scientist John McCarthy, who coined the name "artificial intelligence". The purpose of this conference was to explain how machines could be designed to simulate human intelligence. Early artificial intelligence systems focused on symbolic reasoning and problem solving, but advances in machine learning, data processing, and computing power have evolved AI significantly since then. Some high-profile applications of AI include advanced web search engines (e.g., Google Search); recommendation systems (used by YouTube, Amazon, and Netflix); interacting via human speech (e.g., Google Assistant, Siri, and Alexa); autonomous vehicles (e.g., Waymo); generative and creative tools (e.g., ChatGPT, and AI art); and superhuman play and analysis in strategy games (e.g., chess and Go). However, many AI applications are not perceived as AI: "A lot of cutting edge AI has filtered into general applications, often without being called AI because once something becomes useful enough and common enough it's not labeled AI anymore.

Move towards adopting the use of artificial intelligence is a critical concept in understanding how customers can adopt and interact with AI-based features offered by banks. This goal goes beyond simple knowledge or interests; reflects users' willingness to adopt AI technology as part of their regular banking experience, indicating the extent to which they perceive the value and convenience of AI-based solutions. (5)

2.3 Literature review and research hypotheses

The first study aims to examine the factors influencing the adoption of M-banking from the perspective of developing countries (the case of tunisia) by extending the technology acceptance Model (TAM) . The study was conducted in tunisia .To achieve this structural equation modeling with partial structural equation modeling (PLS-SEM) was used to analyze data from 202 responses . The evidence showed that perceived ease of use and perceived usefulness . Also perceived security determined the intention to adopt M-banking . In this study the empirical that trust partially mediates the relationship between perceived security and satisfaction . Based on the research findings , managerial and theoretical implications are discussed . Thus , the result is m-banking as a new technology is facing many difficulties to spread generally in developing countries and particularly in tunisia . (1)

The Purpose of second study is to examine the important factors that help explain customers' willingness to adopt mobile banking (M-banking). For this purpose, the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) was applied and in order to more accurately predict customers' behavioral intentions, an attempt was made to extend it.The research data was collected from 396 customers of Iranian private banks who had experiences of using M-banking. Structural equation modeling technique was used to test the research hypothesis. The study tasted UTAUT2 and trialability constructs to better understand M-banking adoption in Iran It also evaluated customer behaviors . (2)

And about the third study Mobile banking is today an important and constantly evolving tool for conducting banking transactions. It has great potential in a developing country like India. This study explores the significant antecedents of mobile banking adoption intention of Indian customers and proposes a comprehensive framework by extending the traditional technology acceptance model (TAM). In addition to the two concepts provided by TAM, four customer-oriented concepts were also measured for this purpose. The conceptual model was verified empirically, with the data mobilized with a survey of 203 potential users of mobile banking services. Structural equation modeling (SEM) technique was undertaken to determine the effect of antecedents on mobile banking adoption intention.This study provides meaningful information about the factors affecting mobile banking adoption .(3)

In the four study, For financial institutions , mobile banking has represented a major advance in remote banking services . However , many customers are ensure about its security .This study develops , perceived risk and trust into the classic TAM model to shed light on the factors that determine user acceptance of mobile banking applications . Participants were asked to examine a mobile app from the largest bank in europe . In the proposed model , an external influence approach was included , theoretically and initially stated by (DAVIS et al .1989) . The proposed model was tested empirically using data collected from an online survey with the application of structural equation modeling (SEM) . The result obtained in this study show how attitude mainly determines the intended use of mobile applications rejecting utility and risk factors that directly improve their use . Finally the study shows the main management implications and identifies some strategies to this new activity in the context of new technological advances . (4)

The purpose of the five study is to analyze customer satisfaction of mobile banking services as influenced by security and trust mediated by convenience factors for Livin' da Mandiri users in the Ahmad Yani branch in Tangerang in Indonesia. This study used a quantitative approach. The survey was conducted among 242 Livin' by Mandiri users in Ahmad Yani Agency in Tangerang. This study uses Partial Least Squares SEM or PLS-SEM and identifies the relationship between conceptual and operational variables and definitions using previous research, which includes the use of safety and trust variables as independent variables, customer satisfaction as and dependent variables and comfort factors as mediating variables. The results of this study show that Security has a positive and significant effect on the Convenience Factor.(5)

In six study, Today, mobile banking applications are an integral part of people's lives due to their flexibility and convenience. Despite these advantages, their growth in developing countries is exceeding expectations. However, the use of mobile devices to conduct financial transactions comes with many risks. This article aims to explore the reasons why customers do not use new channels in developing countries, with a particular focus on Nigeria. The study uses two analytical methods, artificial intelligence (AI)-based methods and structural equation modeling (SEM). A feedforward neural network (FFNN) sensitivity screening technique was used to select the most dominant parameters from the mobile banking data collected from 823 respondents. Four algebraic regressors were used to validate the study's AI-based model. The results of the AI study revealed that risk, trust, enabling conditions and inadequate digital laws are the most dominant parameters affecting the growth of mobile banking in Nigeria.(6)

The following Table () provides a summary of the most related previous studies:

Researcher name	The study title	Aim of the study	Place of study	The variables	The result
Hela Ben Abdennebi (2023)	m-banking adoption from the developing countries perspective :A Mediated model	to examine factors that influence from M-banking adoption from developing countries perspective by extending technology acceptance mode	Tunisia	perceived usefulness, user satisfaction, perceived ease of use, perceived security, trust, in mobile banking	M-banking as a new technology facing many difficulties to spread generally in developing countries and particularly in tunisia
Marzieh Sadeghi& Fatemeh Yahyayi Kharkeshi& Hedyeh Ruholahpur & Majid Fattahi (2021)	Extending UTAUT2 and in M-banking adoption actual use behavior:Does WOM communication matter?	The purpose of this study is to investigate important factors that help explain customer willingness to adopt mobile banking	Iran	Mobile banking, WOM, UTAUT2, Actual use behavior, M-banking adoption	The study tested UTAUT2 and trialability constructs to better understand M- banking adoption in iran It also evaluated customer behaviors.
Ashish kumar & Sanjay Dhingra & Vikas Batra & Harish Purohit (2020)	Framework of Mobile Banking Adoption	explores the important antecedents of the mobile banking adoption intention customers and proposes	India	mobile banking adoption intention perceived ease of use perceived usefulness TAM	This study provides meaningful information about the factors affecting mobile banking adoption
F.munoz s.climent F.liebana (2017)	Determinants of intention to use the mobile banking apps:An extension of the	This study develops a technology acceptance model that integrates	Spain	perceived risk, perceived ease of use,	The result obtained in this study demonstrate how attitudes determine mainly the intended use of mobile apps discarded

	classic TAM model	the innovation diffusion theory perceived risk and trust in the classic TAM model in order to shed light on what factors determine user acceptance of mobile banking applications		perceived usefulness intention to use attitude social media	usefulness and risk as factors directly improve its use
Fimi catur wulan suci kandi sofia senastri dahlan (2023)	The effect security and trust on mobile banking customer satisfaction mediated by convenience factors	to analyze mobile banking customer satisfaction influenced by security and trust mediated by convenience factors for living by mandiri users at the Tangerang Ahmad Yani branch	indonesia	security ,trust, customer satisfaction ,convenience factor ,Mobile banking	The result of this study shows every variable has a positive significant effect on other factors
Nadire cavus,Yakubu bala mohammed and Mohammed nasiru yakubu (2021)	An artificial intelligence-based model for prediction of parameters affecting sustainable growth of mobile banking apps	This paper aims to investigate customers reasons for non-usage of the new conduits in developing countries with distinct interest in nigeria	Nigeria	Information quality service quality system quality trust legal framework risk facilitating condition , social influence	The result found risk , facilitating conditions and inadequate digital laws to be the most dominant parameters that affect mobile banking growth discovered social influence service quality to have no influence

Chapter THREE:

Methodology and Design

Chapter THREE

Methodology and Design

This chapter presents the methodology used by this study to measure the extent to which the factors in our model influence the intention to use artificial intelligence in mobile banking. This chapter includes the research design, study sample design, population, data collection, validity and reliability of the study instrument, as well as a description of the study sample .

3.1 Research design

The research design is the method of carrying out the entire research project. The term "study design" refers to "the process of the research methodology to be applied" (Hair, 2010). The research methodology must be consistent with the research objectives and the data needed for the analysis. Multiple designs, including experimental or descriptive methods, can be used to achieve research objectives. Since the present research aims to identify the factors influencing the intention to use artificial intelligence in mobile banking from the perspective of workers in the city of RIYADH, the descriptive/quantitative approach is consistent with this study. Also, quantitative research turns data into numbers and uses mathematical and statistical methods to analyze those numbers. To obtain scientific answers to the research question, data is collected using a (questionnaire).

3.2 Research Methodology

This research used the descriptive, analytical, statistical approach through questionnaire as a tool to collect data. The quantitative method helps in offering real data in the empirical investigation of factors influencing the intention to use artificial intelligence in mobile banking. In this research, a survey was conducted to determine the factors affecting the intention to use artificial intelligence in mobile banking from the point of view of Sattam bin Abdulaziz university student. The survey has been built - based on a comprehensive review of the latest literature related to the research topic.

3.3 Statistical Analysis Report

1. Introduction

This report presents a comprehensive statistical analysis of factors influencing users' intention to adopt AI-powered features in mobile banking. The analysis includes demographic characteristics, user attitudes, perceived benefits, trust factors, experience-related aspects, and their correlations.

1. Demographic Characteristics

Table 1: Participant Demographics

Category	Subgroup	Frequency	Percentage
Gender	Male	68	68%
	Female	32	32%
Age	<20	10	10%
	20-30	70	70%
	30-40	10	10%
	>40	10	10%
Education	High School	20	20%
	Bachelor's	70	70%
	Post graduate	10	10%
Experience	Beginner	15	15%
	Average	40	40%
	Advanced	45	45%

Table 1 presents the demographic characteristics of the study participants. A total of 100 individuals participated, with a gender distribution of 68% male and 32% female.

Regarding age, the majority of respondents (70%) were between 20 and 30 years old, while the remaining participants were equally distributed across the other age categories: less than 20 years (10%), between 30 and 40 years (10%), and over 40 years (10%).

In terms of educational attainment, 70% of the participants held a bachelor's degree, 20% had completed high school, and 10% possessed postgraduate qualifications.

Concerning professional experience, 45% of the participants were categorized as advanced, 40% reported average experience, and 15% were classified as beginners.

These demographic data suggest a sample predominantly composed of young, educated individuals with varying levels of professional experience, which may have implications for the interpretation and generalizability of the study findings.

2. Overall Attitudes Toward AI in Banking

Table 2: General Attitudes with Descriptive Statistics

Statement	Response	Frequency	Percentage	Mean	SD
AI improves financial management	Completely Agree (5)	42	42%	3.98	1.12
	Agree (4)	26	26%		
	Neutral (3)	20	20%		
	Disagree (2)	8	8%		
	Completely Disagree (1)	4	4%		
Positive experience with AI features	Completely Agree (5)	35	35%	3.82	1.08
	Agree (4)	37	37%		
	Neutral (3)	18	18%		
	Disagree (2)	6	6%		
	Completely Disagree (1)	4	4%		
Intention to use AI services	Completely Agree (5)	38	38%	3.91	1.15
	Agree (4)	27	27%		
	Neutral (3)	22	22%		
	Disagree (2)	9	9%		
	Completely Disagree (1)	4	4%		

Table 2 summarizes participants' general attitudes toward AI in financial management, including descriptive statistics for each statement.

Regarding the perception that "AI improves financial management," 42% of respondents completely agreed and 26% agreed, indicating a generally positive sentiment. Only a small proportion expressed disagreement (8%) or complete disagreement (4%). The mean score for this statement was 3.98 (SD = 1.12), suggesting a moderately high level of agreement overall.

Concerning the statement "Positive experience with AI features," 35% of participants completely agreed and 37% agreed. Meanwhile, 18% remained neutral, and a small percentage disagreed (6%) or completely disagreed (4%). The mean score was 3.82 (SD = 1.08), reflecting a generally favorable evaluation of AI features.

As for the "Intention to use AI services," 38% of the respondents completely agreed and 27% agreed. Neutral responses accounted for 22%, while 9% disagreed and 4% completely disagreed. The mean value for this statement was 3.91 (SD = 1.15), highlighting an overall positive intention to adopt AI services.

These findings demonstrate a consistent trend of favorable attitudes toward AI applications, though a minor segment of participants showed reservations, as indicated by the neutral and disagreement responses.

3. Key Factors Analysis

Table 3: Perceived Benefits

Factor	Subgroup	Positive Response (Agree+Completely)	Percentage
Time/Effort Reduction	All Users	65	65%
	Beginners	8	53%
	Advanced	37	82%
Financial Management	Male	53	78%
	Female	15	47%

Table 3 presents the perceived benefits of AI among participants, focusing on time and effort reduction and financial management improvements.

Regarding time and effort reduction, 65% of all users reported positive responses (agree or completely agree). When broken down by experience level, 53% of beginners and 82%

of advanced users indicated positive perceptions, suggesting that more experienced users were significantly more likely to acknowledge the efficiency benefits of AI.

In terms of financial management, gender-based differences were observed. Among male participants, 78% expressed positive perceptions regarding AI's contribution to financial management, compared to only 47% of female participants. This disparity highlights potential variations in perceived value and acceptance of AI applications across gender groups.

Overall, the data reveal that experience level and gender play influential roles in shaping the perceived benefits of AI among users.

Table 4: Trust Factors

Factor	Subgroup	Positive Response	Percentage
Fraud Detection	All Users	68	68%
Data Protection	Male	49	72%
	Female	19	58%
Safety Guarantees	<30 years	42	60%
	>30 years	16	40%

Table 4 outlines participants' perceptions of trust factors associated with AI services, including fraud detection, data protection, and safety guarantees.

Overall, 68% of users positively acknowledged AI's role in fraud detection, indicating a strong level of trust in this capability.

In terms of data protection, gender differences emerged. Among male participants, 72% expressed positive responses, compared to 58% of female participants. This suggests a higher level of confidence in AI data protection mechanisms among males.

Regarding safety guarantees, age appeared to be a relevant factor. Among participants younger than 30 years, 60% reported positive responses, whereas only 40% of those older than 30 years did so. These results suggest that younger users tend to have greater trust in the safety assurances provided by AI systems.

In summary, trust in AI services appears to vary notably according to gender and age, with males and younger participants generally expressing higher levels of trust.

Table 5: Experience Factors

Factor	Subgroup	Positive Response	Percentage
Ease of Use	All Users	72	72%
Interface Quality	High School	13	65%
	Bachelor's	49	70%
	Post graduate	9	90%
Learning Curve	<20 years	6	60%
	20-30 years	53	76%

Table 5 presents the participants' perceptions regarding experience-related factors, including ease of use, interface quality, and the learning curve associated with AI services. Ease of use received positive responses from 72% of all users, indicating a generally favorable evaluation of AI systems in terms of usability.

With regard to interface quality, perceptions varied by educational attainment. Among participants with a high school education, 65% provided positive responses, while 70% of bachelor's degree holders and 90% of postgraduate participants reported satisfaction. This trend suggests that higher educational levels are associated with greater appreciation of interface quality.

Concerning the learning curve, younger participants showed more positive responses. Specifically, 60% of users under 20 years of age responded positively, compared to 76% of users aged between 20 and 30 years. These findings imply that slightly older, yet still young users, may find it easier to adapt to AI technologies.

Overall, the results highlight that educational background and age influence the perceived ease of use, interface quality, and learning experience of AI services.

4. Correlation Matrix with Significance

Table 6: Pearson Correlation Matrix

Variable	1	2	3	4	5
1. Intention to Use	1				
2. Perceived Benefits	.78** (.000)	1			
3. Trust in Security	.82** (.000)	.65** (.001)	1		
4. User Experience	.75** (.000)	.58** (.003)	.71** (.000)	1	
5. Ease of Learning	.63** (.002)	.52* (.012)	.59** (.004)	.67** (.001)	1

Note: Values represent correlation coefficients (p-values in parentheses)

$p < .05$, ** $p < .01$ •

Table 6 displays the Pearson correlation coefficients among the key study variables: intention to use, perceived benefits, trust in security, user experience, and ease of learning.

Intention to use was strongly and positively correlated with perceived benefits ($r = .78$, $p < .01$), trust in security ($r = .82$, $p < .01$), user experience ($r = .75$, $p < .01$), and ease of learning ($r = .63$, $p < .01$). These results suggest that higher perceptions of benefits, greater trust in security, better user experience, and easier learning processes are all significantly associated with a stronger intention to adopt AI services.

Perceived benefits were also positively correlated with trust in security ($r = .65$, $p < .01$), user experience ($r = .58$, $p < .01$), and ease of learning ($r = .52$, $p < .05$). Similarly, trust in security showed significant positive correlations with user experience ($r = .71$, $p < .01$) and ease of learning ($r = .59$, $p < .01$). Additionally, user experience was positively related to ease of learning ($r = .67$, $p < .01$).

Overall, the correlation matrix indicates a cohesive and mutually reinforcing relationship among the constructs, with particularly strong associations involving trust in security and perceived benefits.

5. Demographic Comparisons

Table 7: Intention to Use by Demographics

Demographic	Positive Intention	Frequency	Percentage
Age	<20	6	60%
	20-30	49	70%
	30-40	5	50%
	>40	4	40%
Education	High School	12	60%
	Bachelor's	47	67%
	Postgraduate	8	80%
Gender	Male	47	69%
	Female	18	56%

Table 7 summarizes participants' positive intention to use AI services across different demographic groups, including age, education level, and gender.

In terms of age, participants aged 20–30 years demonstrated the highest level of positive intention (70%), followed by those under 20 years (60%), participants aged 30–40 years (50%), and those over 40 years (40%). These results indicate that younger participants, particularly those between 20 and 30 years old, are more inclined toward adopting AI services.

Regarding educational attainment, postgraduate participants exhibited the highest positive intention to use AI services (80%), compared to 67% of bachelor's degree holders and 60% of those with a high school education. This suggests that higher education levels are associated with a greater willingness to engage with AI technologies.

Gender differences were also observed: 69% of male participants expressed a positive intention to use AI services, compared to 56% of female participants. This finding points to a higher adoption intention among male users.

Overall, the results suggest that younger, more highly educated, and male participants are more likely to have a positive intention toward using AI services.

6. Relative Impact of Factors

Table 8: Factor Contribution to Adoption Intention

Factor Group	Percentage Contribution
Trust Factors	40%
User Experience	35%
Perceived Benefits	25%

Table 8 presents the contribution of different factor groups to participants' intention to adopt AI services.

Trust factors emerged as the most significant contributors, accounting for 40% of the variance in adoption intention. This highlights the critical role of perceived security, fraud detection, and data protection in influencing users' decisions to engage with AI technologies.

User experience factors contributed 35% to adoption intention, indicating that the ease of use, interface quality, and learning curve are substantial drivers of user willingness to adopt AI services.

Perceived benefits contributed 25%, suggesting that while the advantages offered by AI, such as time and effort reduction and improved financial management, are important, trust and user experience play comparatively larger roles in shaping adoption behavior. Overall, the findings emphasize that building trust and ensuring a positive user experience are key strategies for enhancing AI service adoption.

7. Chi-Square Tests for Independence

Table 9: Chi-Square Test Results

Relationship Tested	χ^2 Value	df	p-value	Significant?
Experience Level × Intention to Use	15.72	4	0.003	Yes**
Education × Trust in Fraud Detection	12.35	4	0.015	Yes*
Gender × Data Protection Concerns	9.84	2	0.043	Yes*
Age Group × Perceived Time Savings	8.26	6	0.219	No

$p < .05$, ** $p < .01$

Table 9 presents the results of the chi-square tests conducted to examine associations between key demographic variables and AI adoption-related factors.

A significant association was found between experience level and intention to use AI services ($\chi^2(4) = 15.72$, $p = .003$), indicating that users' experience levels significantly influence their adoption intentions. Similarly, a significant relationship was observed between education level and trust in fraud detection ($\chi^2(4) = 12.35$, $p = .015$), suggesting that higher educational attainment may be linked to greater trust in AI's security features.

Gender differences in concerns over data protection were also statistically significant ($\chi^2(2) = 9.84$, $p = .043$), highlighting that gender may play a role in shaping perceptions of data security risks.

However, the association between age group and perceived time savings was not significant ($\chi^2(6) = 8.26$, $p = .219$), indicating that perceptions of AI's time-saving benefits do not vary significantly across age groups.

These results suggest that experience, education, and gender are important demographic factors influencing user trust and adoption of AI services, whereas age appears less influential in perceived benefits related to time efficiency.

First: Data-Based Conclusion

Positive Attitudes Generally Regarding AI:

- With a mean score of 3.98 out of 5, the majority of participants concurred that AI enhances financial management.
- Likewise, there was a high inclination to adopt AI services (mean score of 3.91).

Important Elements Affecting Intention to Use:

- Trust in Security (Trust Factors): The largest contributing element, making up 40% of the total.
- User Experience: This component is also quite important, accounting for 35%.
- Perceived ease of use: Significant, but at 25%, not as significant.

Differences in Demographics:

- Age: Participants between the ages of 20 and 30 had the highest intention to use (70%).
- Education: The greatest intention was shown by postgraduate participants (80%).
- Gender: Compared to women, men were more likely to employ AI (69% vs. 56%).

Further Examination Using Statistical Tests: Experience level and intention to use were shown to be significantly correlated.

- The degree of education and confidence in fraud detection.
- Gender and data security concerns.

1. Why Was Security Trust the Most Vital Factor?

- Transactions involving money are inherently delicate.
 - People must have faith that their financial information won't be stolen or used improperly.
- Users are therefore more inclined to embrace AI technologies if they believe they provide robust protection (fraud detection, data security).

2. Why Did User Experience Have Such an Impact?

- AI applications in banking are still in their infancy.
- The interface must be simple, responsive, and straightforward for users to grasp.
- Even if a system is safe and helpful, consumers could be reluctant to utilize it if it is hard to use.

3. Why Did Perceived Benefit Have Less of an Effect? • The majority of users put security and usability above additional benefits, even though time savings and better money management are significant. Put another way, people will steer clear of an app if it is helpful yet unsecure or hard to use.

4. What Makes Users Who Are Younger and Better Off More Accepting? • Younger generations (20–30) are less resistant to change and more able to adjust to technology. • Users who have more education are better able to comprehend AI systems and see their benefits since they have improved analytical abilities.

5. Why Did Men Display Greater Degrees of Trust? • Social studies show that men are typically more eager to take chances when experimenting with new technology, particularly those that haven't been thoroughly tested yet. Men might also feel more in charge of security systems than women do.

6. Why Did Age Not Have a Significant Impact on Perceptions of Time-Saving? • People of all ages acknowledged that AI saves time. • Since everyone, regardless of age, values saving time, there was no discernible difference.

Final Summary

- Trust and security come first, followed by user experience, then perceived benefits.
- Younger, more educated users are more inclined to adopt AI services, and males tend to trust AI security more than females.
- The success of promoting AI banking services heavily depends on building trust and ensuring ease of use.

Chapter FOUR:

Closure

Chapter FOUR

Closure

4.1 conclusion

Conclusion in conclusion, this study the main factors that influence the purpose of using artificial intelligence (AI) in furniture bank services. Search has focused on some key elements as confidence, user satisfaction, perceive to use, and the real user's behavior. Throughout the project, became clear that these factors play an important role in promoting or harnessing acrovie in bathtubs. Faith appeared As a critical item, as users have to be convinced that systems will project their data, maintain reliable security. Without strong trust, unlikely that users to engage with the functions of this, as advanced or useful. Always, the user satisfaction is identified as essential; If the bank services experience you are committed to it or overcome the customer expectations, users will be discouraged to approve these long -term funds. perceived The use facility also attempted to be very influential. Users usually prefer intuitive systems, simple and they need no major efforts important. Institutions of consequence, institutions must be in designation of the solutions of the solutions of one who are friendly, easy to navigate and transpose in evil platform. In addition, understanding the relationship between the current goal and behavior was required because you only find positive goals are not enough; The real success If you assume to the processing of these goals in regular use models. After analysis analysis several prior and shrugged shrinking studies is landing a beautiful ebications, some challenges should be answered. These include assuring the confidential data, minimizing perceived, educated in the perks of her, and the improbable of the user's global.

4.2 recommendation

from my point of view, the most vital factor between all Put the strong trust between users and systems of it. The confidence is the basis of all the following - the user satisfaction, the desire to try new functions and, after all, loyalty to the service. As a consequence, financial institutions must prioritare transparent communication, strong security measures and the quality of consistent service to promote the confidence. In the bank, for the banker seeking for certain a competition creeping and practice it but a necessity, but a need but needy but be needed but untoity. While the address The main accussces, they can unlock artificial intelligence, provide more customer experiences, and the position as leaders to the quick ratio.

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استكمال لما سبق تم جمع بعض العينات لدراسة أبعاد هذا البحث عن طريق تقديم استبانة تقيس مدى تأثير نية استخدام الذكاء الاصطناعي في الخدمات المصرفية عبر الهاتف المحمول لدى طلاب وطالبات جامعة الأمير سطام :

نحن طلاب جامعة الأمير سطام بن عبدالعزيز ، كلية إدارة الأعمال ، تخصص نظم المعلومات الادارية ، لإتمام مقرر مشروع التخرج نقوم بإجراء دراسة عن العوامل المؤثرة في نية استخدام الذكاء الاصطناعي للعملاء بالخدمات المصرفية عبر الهاتف المحمول ونضع بين يديكم هذا الاستبيان الذي يهدف إلى جمع بيانات تسهم في فهم هذه العوامل

نود منكم التكرم مشكورين بالمشاركة في هذا الاستبيان الذي لن يستغرق سوى عدة دقائق من وقتكم الثمين . و نؤكد لكم أن جميع البيانات التي يتم جمعها ستعامل بسرية تامة ولن تُستخدم إلا لأغراض هذا المشروع

الجزء الأول: المعلومات الديمغرافية

الجنس : ذكر () انثى ()

العمر : أقل من 20 () من 20 إلى 30 () من 30 إلى 40 () أكثر من 40 ()

المستوى التعليمي : الثانوي أو أقل () بكالوريوس () دراسات عليا ()

الجزء الثاني : مقياس مستوى الخبرة

مستوى خبرتك في استخدام التطبيقات المصرفية : مبتدئ () متوسط () متقدم ()

الجزء الثالث : اسئلة مبنية على موضوع وعوامل البحث

يرجى وضع علامة (x) تحت الخيار الذي يتفق مع رأيك في العبارات التالية :

العوامل المؤثرة على نية استخدام الذكاء الاصطناعي في الخدمات المصرفية عبر الهاتف المحمول

الرقم	العبارة	موافق بشدة	موافق	محايد	لا أوافق	لا أوافق بشدة
١	هل تعتقد أن الذكاء الاصطناعي سيساعدك في تحسين إدارتك المالية، مما يشجعك على استخدامه					
٢	استخدم الخدمات المصرفية التي تعتمد على الذكاء الاصطناعي بكثرة مقارنة بالطرق التقليدية					
٣	أستخدم الخدمات المصرفية عبر الهاتف بشكل منتظم					
٤	لدي نية قوية لاستخدام الذكاء الاصطناعي في الخدمات المصرفية مستقبلاً					

الثقة

الرقم	العبارة	موافق بشدة	موافق	محايد	لا أوافق	لا أوافق بشدة
١	أشعر بالثقة والرضا عند استخدام الذكاء الاصطناعي في الخدمات المصرفية					
٢	أعتقد أن الذكاء الاصطناعي يحمي بياناتي الشخصية بشكل جيد					
٣	أعتقد أن البنوك تقدم ضمانات كافية لجعل الخدمات المصرفية المعتمدة على الذكاء الاصطناعي آمنة وموثوقة					
٤	بمدى عالي اثق في قدرة الذكاء الاصطناعي على اكتشاف الاحتيال وحماية حساباتك المصرفية؟					

رضاء المستخدم

١	أجد أن هذه الخدمات توفر لي تجربة إيجابية				
٢	أشعر بالراحة عند الاعتماد على الذكاء الاصطناعي في معاملاتي اليومية				
٣	تفي الخدمات المصرفية المعتمدة على الذكاء الاصطناعي بتوقعاتي				
٤	أعتقد أن الذكاء الاصطناعي سيحسن تجربتي المصرفية				

سهولة الاستخدام الملحوظة

١	إلى مدى عالي جدا اعتقد أن استخدام الذكاء الاصطناعي في تطبيقات البنوك يقلل من الوقت والجهد اللازمين لتنفيذ العمليات المصرفية				
٢	واجهة المستخدم في هذه الخدمات مريحة وسهلة التنقل				
٣	اعتقد أن الذكاء الاصطناعي في تطبيقات البنوك يوفر تجربة استخدام بديهية حتى لمن ليس لديهم خبرة تقنية				

٤	من السهل تعلم كيفية استخدام الميزات المدعومه بالذكاء الاصطناعي في التطبيقات المصرفيه				
سلوك الاستخدام					
١	أميل الى تجربة ميزات جديدة تعتمد على الذكاء الاصطناعي				
٢	سأوصي الآخرين باستخدام الخدمات المصرفية التي تعتمد على الذكاء الاصطناعي				
٣	تؤثر تجربتي السابقة مع الذكاء الاصطناعي في التطبيقات المصرفية على قراري بالاستمرار في استخدامها مستقبل				
٤	اعتمد بشكل كبير على ميزات الذكاء الاصطناعي، مثل المساعدات الافتراضية أو التحليل المالي الذكي، عند استخدام التطبيق المصرفي				

In continuation of the above, some samples were collected to study the dimensions of this research by presenting a questionnaire that measures the extent of the impact of the intention to use artificial intelligence in mobile banking services among male and female students at Prince Sattam University:

We are students at Prince Sattam bin Abdulaziz University, College of Business Administration, majoring in Management Information Systems. To complete our graduation project, we are conducting a study on the factors influencing customers' intention to use artificial intelligence in mobile banking services.

We present to you this survey, which aims to collect data to help understand these factors.

We would like to thank you for participating in this survey, which will only take a few minutes of your valuable time.

We assure you that all data collected will be treated with strict confidentiality and will be used only for the purposes of this project.

Part One: Demographic Information

Gender: Male () Female ()

Age: Under 20 () 20-30 () 30-40 () Over 40 ()

Educational Level: High School or Below () Bachelor's () Postgraduate ()

Part Two: Experience Level Scale Your level of experience using banking applications:

Beginner () Intermediate () Advanced ()

Part Three: Questions based on the topic and research factors

Please mark (x) under the option that agrees with your opinion on the following statements:

Factors affecting the intention to use artificial intelligence in mobile banking

No.	Items	Strongly agree	Agree	Neutral	Disagree	Strongly disagree
1	Do you think that artificial intelligence will help you improve your financial management, which would encourage you to use it?					
2	I use AI-based banking services more frequently than traditional methods.					
3	I use mobile banking regularly.					
4	I have a strong intention to use artificial intelligence in banking services in the future.					

Trust						
1	I feel confident and satisfied with the use of artificial intelligence in banking services.					
2	I think AI protects my personal data well.					
3	I believe banks provide sufficient guarantees to make AI-based banking safe and reliable.					
4	How confident are you in AI's ability to detect fraud and protect your bank accounts?					
User satisfaction						
1	I find these services provide me with a positive experience.					
2	I feel comfortable relying on AI in my daily transactions.					
3	AI-powered banking meets my expectations.					

4	I believe AI will improve my banking experience.					
Perceived ease of use						
1	"To a very high extent, I believe that the use of artificial intelligence in banking applications reduces the time and effort required to carry out banking operations."					
2	The user interface in these services is convenient and easy to navigate.					
3	I believe that AI in banking applications provides an intuitive user experience even for those without technical expertise.					

4	It's easy to learn how to use AI-powered features in banking apps.					
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Use behavior						
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1	I tend to try new features based on artificial intelligence.					
2	I will recommend others to use banking services that rely on artificial intelligence					
3	My previous experience with AI in banking applications influences my decision to continue using it in the future.					

4	I heavily rely on AI features, such as virtual assistants or smart financial analysis, when using the banking application.					
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